

Post-Training

- The details of post-training are often undisclosed by the companies that develop these models
- However, it is known that the post-training process has an initial step where the “core skills” are defined, and that the acquisition of these skills is achieved using these two components:
 1. Supervised fine-tuning (SFT)
 2. Reinforcement learning using policy gradient methods*

* There are also newer methods that use other algorithms

Core Skills in TüLU 3



- Other models may share or diverge from this set
- Curation of prompts for these "core skills" is a combination of data curation and synthesis

Supervised Fine-Tuning (SFT)

- The simplest training method employed
- Identical to fine-tuning a model for any other task
- Training proceeds by feeding the decoder with the prompt and training it to predict the first word,
 - then the prompt and the first word, training it to predict the second
 - then the prompt and the first two words, training it to predict the third
 - ...
- Main shortcomings for this method: exposure bias and inability to learn from negative examples

Results

- As mentioned last class: very difficult to evaluate
 - Many benchmarks used, testing for a wide variety of capabilities

Training for MATH does not transfer well to Deepmind Math. One difference between MATH, which is part of our development evaluations and DeepMind Math, which is in our unseen evaluations is that the former often

Models generally overfit to IFEval. We find that there is a significant difference between performance on IFEval and IFEval-OOD of all the models, even though we created the latter to be structured very similar to the original dataset, just with a disjoint set of constraints. We observe that instruction following with verifiable constraints is a

Instruction following performance varies across categories. We observe that the relative performance of TüLU 3 models on AlpacaEval is different from that on HREF. This may be explained by the fact that instruction following is a highly diverse task, and the distributions of HREF and AlpacaEval may differ, with some categories of instructions